

<i>termvar</i> , x	term variable
<i>label</i> , l , k	field label
<i>index</i> , i , j , n , m	indices

T, S, U	$::=$ $ $ Top $ $ $T \rightarrow T'$ $ $ $\{l_1 : T_1, \dots, l_n : T_n\}$ $ $ (T)	 M	type maximum type type of functions record
t	$::=$ $ $ x $ $ $\lambda x : T. t$ $ $ $t t'$ $ $ $\{l_1=t_1, \dots, l_n=t_n\}$ $ $ $t.l$ $ $ let $p=t$ in t' $ $ (t) $ $ $[x \mapsto t]t'$ $ $ σt	 bind x in t bind $b(p)$ in t' M M M	term variable abstraction application record projection pattern binding
p	$::=$ $ $ $x : T$ $ $ $\{l_1=p_1, \dots, l_n=p_n\}$	 $b = x$ $b = b(p_1..p_n)$	pattern variable pattern record pattern
v	$::=$ $ $ $\lambda x : T. t$ $ $ $\{l_1=v_1, \dots, l_n=v_n\}$	 bind x in t 	values abstraction record
Γ, Δ	$::=$ $ $ empty $ $ $\Gamma, x : T$ $ $ $\Gamma_1, \dots, \Gamma_n$	 M	type environment
σ	$::=$ $ $ $[x \mapsto t]$ $ $ $\sigma_1, \dots, \sigma_n$		multiple term substitution
<i>terminals</i>	$::=$ $ $ λ $ $ $\rightarrow$ $ $ $\Rightarrow$ $ $ $\vdash$ $ $ $\longrightarrow$ $ $ $\forall$ $ $ $<:$ $ $ $\mapsto$ $ $ $\wedge$ $ $ $\vee$ $ $ $=$		
<i>formula</i>	$::=$ $ $ <i>judgement</i>		

		$x=x'$	
		$(formula)$	
		$\neg formula$	
		$\forall i \in 1..m. formula$	
		$\exists i \in 1..m. formula$	
		$formula \wedge formula'$	
		$l=l'$	
		$formula_1 \quad \dots \quad formula_n$	
$Judgement_in$	$::=$		
		$x \in \mathbf{dom}(\Gamma)$	
		$x : T \in \Gamma$	
$Jtype$	$::=$		
		$\Gamma \vdash \mathbf{ok}$	type environment Γ is well-formed
		$\Gamma \vdash T$	type T is well-formed in type environment Γ
		$\Gamma \vdash S <: T$	S is a subtype of T
		$\Gamma \vdash t : T$	term t has type T
		$\vdash p : T \Rightarrow \Delta$	pattern p matches type T giving bindings Δ
Jop	$::=$		
		$t_1 \longrightarrow t_2$	t_1 reduces to t_2
		$\mathbf{match}(p, v)=\sigma$	
$judgement$	$::=$		
		$Judgement_in$	
		$Jtype$	
		Jop	
$user_syntax$	$::=$		
		$termvar$	
		$label$	
		$index$	
		T	
		t	
		p	
		v	
		Γ	
		σ	
		$terminals$	
		$formula$	

$$\boxed{x \in \mathbf{dom}(\Gamma)}$$

$$\frac{}{x \in \mathbf{dom}(\Gamma, x : T)} \quad \text{XING_1}$$

$$\frac{x \in \mathbf{dom}(\Gamma)}{x \in \mathbf{dom}(\Gamma, x' : T')} \quad \text{XING_3}$$

$$\boxed{x : T \in \Gamma}$$

$$\frac{}{x : T \in \Gamma, x : T} \quad \text{TIN_1}$$

$$\frac{x : T \in \Gamma}{x : T \in \Gamma, x' : T'} \quad \text{TIN_3}$$

$\boxed{\Gamma \vdash \mathbf{ok}}$ type environment Γ is well-formed

$$\frac{}{\mathbf{empty} \vdash \mathbf{ok}} \quad \text{GOK_1}$$

$$\frac{\Gamma \vdash T \quad \neg(x \in \mathbf{dom}(\Gamma))}{\Gamma, x : T \vdash \mathbf{ok}} \quad \text{GOK_2}$$

$\boxed{\Gamma \vdash T}$ type T is well-formed in type environment Γ

$$\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash \mathbf{Top}} \quad \text{GT_TOP}$$

$$\frac{\Gamma \vdash T \quad \Gamma \vdash T'}{\Gamma \vdash T \rightarrow T'} \quad \text{GT_FUN}$$

$$\frac{\Gamma \vdash T_1 \quad \dots \quad \Gamma \vdash T_n}{\Gamma \vdash \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{GT_RCD}$$

$\boxed{\Gamma \vdash S <: T}$ S is a subtype of T

$$\frac{\Gamma \vdash \mathbf{ok}}{\Gamma \vdash S <: \mathbf{Top}} \quad \text{SA_TOP}$$

$$\frac{\Gamma \vdash T_1 <: S_1 \quad \Gamma \vdash S_2 <: T_2}{\Gamma \vdash S_1 \rightarrow S_2 <: T_1 \rightarrow T_2} \quad \text{SA_ARROW}$$

$$\frac{\forall i \in 1..m. \exists j \in 1..n. (k_i = l_j \wedge \Gamma \vdash S_i <: T_j)}{\Gamma \vdash \{k_1 : S_1, \dots, k_m : S_m\} <: \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{SA_RCD}$$

$\boxed{\Gamma \vdash t : T}$ term t has type T

$$\frac{\Gamma \vdash \mathbf{ok} \quad x : T \in \Gamma}{\Gamma \vdash x : T} \quad \text{TY_VAR}$$

$$\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2} \quad \text{TY_ABS}$$

$$\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}} \quad \text{TY_APP}$$

$$\frac{\Gamma \vdash t_1 : T_1 \quad \vdash p : T_1 \Rightarrow \Delta \quad \Gamma, \Delta \vdash t_2 : T_2}{\Gamma \vdash \mathbf{let } p = t_1 \mathbf{ in } t_2 : T_2} \quad \text{TY_LET}$$

$$\frac{\Gamma \vdash t_1 : T_1 \quad \dots \quad \Gamma \vdash t_n : T_n}{\Gamma \vdash \{l_1 = t_1, \dots, l_n = t_n\} : \{l_1 : T_1, \dots, l_n : T_n\}} \quad \text{TY_RCD}$$

$$\frac{\Gamma \vdash t : \{l_1 : T_1, \dots, l_n : T_n\}}{\Gamma \vdash t.l_j : T_j} \quad \text{TY_PROJ}$$

$$\frac{\Gamma \vdash t : S \quad \Gamma \vdash S <: T}{\Gamma \vdash t : T} \quad \text{TY_SUB}$$

$\boxed{\vdash p : T \Rightarrow \Delta}$ pattern p matches type T giving bindings Δ

$$\overline{\vdash x : T : T \Rightarrow \mathbf{empty}, x : T} \quad \text{PAT_VAR}$$

$$\frac{\vdash p_1 : T_1 \Rightarrow \Delta_1 \quad \dots \quad \vdash p_n : T_n \Rightarrow \Delta_n}{\vdash \{l_1=p_1, \dots, l_n=p_n\} : \{l_1 : T_1, \dots, l_n : T_n\} \Rightarrow \Delta_1, \dots, \Delta_n} \quad \text{PAT_RCD}$$

$\boxed{t_1 \longrightarrow t_2}$ t_1 reduces to t_2

$$\overline{(\lambda x : T_{11}.t_{12}) v_2 \longrightarrow [x \mapsto v_2]t_{12}} \quad \text{REDUCE_APPABS}$$

$$\frac{\mathbf{match}(p, v_1)=\sigma}{\mathbf{let } p=v_1 \mathbf{ in } t_2 \longrightarrow \sigma t_2} \quad \text{REDUCE_LETV}$$

$$\overline{\{l'_1=v_1, \dots, l'_n=v_n\}.l'_j \longrightarrow v_j} \quad \text{REDUCE_PROJRCD}$$

$$\frac{t_1 \longrightarrow t'_1}{t_1 t \longrightarrow t'_1 t} \quad \text{REDUCE_CTX_APP_FUN}$$

$$\frac{t_1 \longrightarrow t'_1}{v t_1 \longrightarrow v t'_1} \quad \text{REDUCE_CTX_APP_ARG}$$

$$\overline{t \longrightarrow t'} \quad \text{REDUCE_CTX_RECORD}$$

$$\frac{}{\{l_1=v_1, \dots, l_m=v_m, l=t, l'_1=t'_1, \dots, l'_n=t'_n\} \longrightarrow \{l_1=v_1, \dots, l_m=v_m, l=t', l'_1=t'_1, \dots, l'_n=t'_n\}}$$

$$\frac{t_1 \longrightarrow t'_1}{\mathbf{let } p=t_1 \mathbf{ in } t_2 \longrightarrow \mathbf{let } p=t'_1 \mathbf{ in } t_2} \quad \text{REDUCE_CTX_LET_BINDING}$$

$\boxed{\mathbf{match}(p, v)=\sigma}$

$$\overline{\mathbf{match}(x : T, v)=[x \mapsto v]} \quad \text{M_VAR}$$

$$\frac{\forall i \in 1..m. \exists j \in 1..n. (l_i=k_j \wedge \mathbf{match}(p_i, v_j)=\sigma_i)}{\mathbf{match}(\{l_1=p_1, \dots, l_m=p_m\}, \{k_1=v_1, \dots, k_n=v_n\})=\sigma_1, \dots, \sigma_m} \quad \text{M_RCD}$$

Definition rules: 30 good 0 bad

Definition rule clauses: 61 good 0 bad